

MTH 346/646 homework cover sheet

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Homework number: 10

Date: November 8, 2025

What topics did this assignment cover? How comfortable do you feel with each of those topics?

This covered the homomorphisms between a curve and its “friend” as well as finding the ranks of curves using methods from the proof of Mordell’s theorem. The former concept is easy enough from the definitions. I feel like I have a surface-level understanding of the ideas around ranks, but I’m still lacking in depth surrounding what’s actually going on with $C(\mathbb{Q})/2C(\mathbb{Q})$.

In the course of doing this assignment, where did you get stuck? How did you get unstuck?

I did a lot of reading the book to try to understand the theory surrounding finding the rank.

Is there anything you’ve done here that you’re not confident about, that you want me to take a really close look at, or that you want me to suggest an alternate approach to?

As you’ll see, I wasn’t really sure how to cap off problems 2 and 3.

1. Let $C : y^2 = x^3 - 2x^2 - 267x$ and $\bar{C} : y^2 = x^3 + 4x^2 + 1072x$ and let $\phi : C \rightarrow \bar{C}$ be the homomorphism described in class.

Let $\bar{P} = (173889/4900, 101618313/343000) \in \bar{C}(\mathbb{Q})$. Find a point $P \in C(\mathbb{Q})$ so that $\phi(P) = \bar{P}$.

Proof. I started by writing down the equations

$$\frac{y^2}{x^2} = \frac{173889}{4900}$$

and

$$\frac{y(x^2 + 267)}{x^2} = \frac{101618313}{343000}.$$

Squaring the second equation and moving things around in both so that you get a $y^2 =$ in each allows us to combine the equations to get

$$\frac{173889}{4900}(x^3 + 267x)^2 = \left(\frac{101618313}{343000}\right)^2 x^4$$

after some further simplifications. I plugged this directly into Wolfram Alpha and got a few solutions for x . Not all give a real value for y after plugging it into C , but one that does gives the point

$$P = \left(-\frac{300}{49}, \frac{12510}{343}\right).$$

I checked that this does indeed map back to $\bar{P}$ as desired. □

2. Compute the rank of the elliptic curve $C : y^2 = x^3 + 73x$. Give representatives for the cosets of $2C(\mathbb{Q})$ in $C(\mathbb{Q})$.

It is OK to use Magma to check your answer, but none of your justification can rely on Magma computations.

Proof. We consider the divisors of 73, which are 1, -1 , 73, and -73 . Since $a = 0$, the equations corresponding to these are

$$\begin{aligned} \text{(i)} \quad N^2 &= M^4 + 73e^4 \\ \text{(ii)} \quad N^2 &= -M^4 - 73e^4 \\ \text{(iii)} \quad N^2 &= 73M^4 + e^4 \\ \text{(iv)} \quad N^2 &= -73M^4 - e^4. \end{aligned}$$

Clearly (ii) and (iv) can have no solutions since both would have a nonnegative LHS and negative RHS (since $M > 0$). The solution $(M, e, N) = (1, 0, 1)$ holds for (i), and a far less obvious solution to (iii) is $(1, 6, 37)$. I found the latter by setting $M = 1$ and trying low values of e . Thus both (i) and (iii) have solutions, so we conclude that $\alpha(C(\mathbb{Q}))$ has order 2.

Now we do the same for $\bar{C} : y^2 = x^3 - 292x$. The divisors of 292 are ± 1 , ± 2 , ± 4 , ± 73 , ± 146 , and ± 292 . Considering just square-free divisors, we narrow this down to ± 1 , ± 2 , ± 73 ,

and ± 146 . We will cheat a little bit (not really) by omitting the negatives at the start, since as before, they will not yield solutions. This leaves us with just four equations to check:

$$\begin{aligned} \text{(i)} \quad N^2 &= M^4 + 292e^4 \\ \text{(ii)} \quad N^2 &= 2M^4 + 146e^4 \\ \text{(iii)} \quad N^2 &= 73M^4 + 4e^4 \\ \text{(iv)} \quad N^2 &= 146M^4 + 2e^4. \end{aligned}$$

The first equation again has the solution $(M, e, N) = (1, 0, 1)$. Using some JavaScript, I was able to find the solution $(1786, 9368, 1060410750)$ for equation (ii). Similarly, we have the solutions $(40, 213, 91762)$ for (iii) and $(3130, 3795, 120115807)$ for (iv). This tells us that the order of $\bar{\alpha}(\bar{C}(\mathbb{Q}))$ is 4. Combining this with the previous result, we get that

$$2^r = \frac{2 \cdot 4}{4} = 2,$$

where r is the rank of C , so C has rank 1.

I'm not quite sure what to do to find representatives for $C(\mathbb{Q})/2C(\mathbb{Q})$. Using the equations

$$x = \frac{b_1 M^2}{e^2}, y = \frac{b_1 M N}{e^3}$$

with our values for b_1 , M , e , and N from the previous two paragraphs, we could find points on the curve, but that only gives us 6 possibilities, and we need 8. $\square$

3. * Prove that if $p \equiv 3 \pmod{8}$, then $C : y^2 = x^3 - p^2 x$ has rank zero. (You will want to use the fact that if $p \equiv 3 \pmod{4}$ is a prime, there are no integer solutions to the congruence $z^2 \equiv -1 \pmod{p}$.)

Proof. I'm assuming p is prime. The square-free divisors of $-p^2$ are ± 1 and $\pm p$. Thus we look at the equations

$$\begin{aligned} \text{(i)} \quad N^2 &= M^4 - p^2 e^4 \\ \text{(ii)} \quad N^2 &= -M^4 + p^2 e^4 \\ \text{(iii)} \quad N^2 &= pM^4 - pe^4 \\ \text{(iv)} \quad N^2 &= -pM^4 + pe^4. \end{aligned}$$

We find that $(M, e, N) = (1, 0, 1)$ is a solution to (i) and $(1, 1, 0)$ is a solution to both (iii) and (iv). Since the total number of equations with solutions here must be a power of 2, we get (ii) for free and conclude that $\#\alpha(C(\mathbb{Q})) = 4$.

Now $\bar{C}$ is $y^2 = x^3 + 4p^2 x$. The square-free divisors of $4p^2$ are ± 1 , ± 2 , $\pm p$, and $\pm 2p$. As in problem 2, negatives will give us no solutions, so we check

$$\begin{aligned} \text{(i)} \quad N^2 &= M^4 + 4p^2 e^4 \\ \text{(ii)} \quad N^2 &= 2M^4 + 2p^2 e^4 \\ \text{(iii)} \quad N^2 &= pM^4 + 4pe^4 \\ \text{(iv)} \quad N^2 &= 2pM^4 + 2pe^4. \end{aligned}$$

Again $(1, 0, 1)$ is a solution to (i). I'm not seeing how I can show that the others have no solutions. I've tried considering them mod 4 and mod p (especially the latter given the hint), but I haven't found the right combination of facts that will help me get there.

If I could show that (ii), (iii), and (iv) had no solutions, then we could conclude that $2^r = 4 \cdot 1/4 = 1$, so the rank $r = 0$. □